

Sima Jamali

Curriculum Vitae

San Diego, USA

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Objective

Highly skilled Software Engineer and Data Scientist with 9 years of experience in developing algorithms and data analysis using state of the art Machine Learning and Optimization to solve complex real-world problems.

Work Experience and Projects

2022–2025 **University of California, San Diego**, Postdoctoral Researcher, Halicioğlu Data Science Institute.

Awarded a Postdoctoral Fellowship to conduct research in machine learning and data science with a focus on recommendation system algorithms for large-scale datasets.

Project: Location-Aware Restaurant Recommendation on Google Maps Data.

Developed an advanced algorithm to enhance restaurant recommendations by leveraging matrix factorization models, prioritizing location relevance with an emphasis on model accuracy and fairness. **(Python, TensorFlow)**

- o Analyzed geographical data to assess the impact of location on the recommended restaurants, uncovering trends and patterns in user preferences.
- o Developed and implemented a framework aimed at increasing locality in recommendations, improving the relevance of suggestions based on user proximity.
- o Investigated popularity bias in recommender systems to enhance fairness.

2015–2021 **Simon Fraser University, Canada**, Research Assistant.

Worked as a researcher at the Computational Logic Lab with a focus on enhancing SAT solvers for industrial optimization problems by leveraging their structural features. Mentored students and supervised research projects related to solvers and their applications in optimization.

Project: Leveraging SAT Instance Structure to Enhance CDCL Solver Performance.

Developed and implemented a novel learnt clause deletion scheme for CDCL SAT solvers based on the properties of industrial SAT instances. **(C++, Python, Gephi)**

- o Developed a novel clause quality measure leveraging the structural properties of SAT formulas.
- o Utilized tools and libraries like Gephi and Graphviz to visualize and analyze graphs from SAT formulas and created a tool to extract advanced features.
- o Integrated innovative heuristics into advanced SAT solvers, enhancing performance.

Project: Innovative Clause Database Management Strategy for SAT Solvers.

Conducted in-depth analysis of learnt clauses in CDCL solvers and Developed advanced techniques in clause database management, leading to enhanced solver efficiency. **(C++)**

- o Implemented the 2nd place SAT solver in SAT competitions 2021.
- o Analyzed learnt clauses in CDCL SAT solvers using various criteria to assess their significance in conflict analysis.
- o Developed and implemented a novel clause management strategy for solvers.

Project: A Crowdsourcing Interface for Object Localization.

Leveraged crowd power for object localization tasks. **(Python, PYBOSSA)**

- o Implemented a platform to use crowdsourcing techniques in object detection and created a new object database using the results for future tasks.

Education

2015–2021 **Simon Fraser University, Vancouver, Canada.**

Ph.D. Computer Science

- o Cumulative GPA: 4.00/4.33

2012–2015 **Sharif University of Technology, Tehran, Iran.**

M.Sc. Computer Engineering - Artificial Intelligence

- o Cumulative GPA: 16.95/20

2008–2012 **Azad University of Mashhad, Mashhad, Iran.**

B.Sc. Information Technology Engineering

- o Cumulative GPA: 17.70/20

Technical Skills

Languages C++, Python, SQL, R.

Tools & Databases Matlab, Git, SQLite, Jupyter Notebook, Gephi.

Libraries NumPy, TensorFlow, Matplotlib, Graphviz.

Selected Publications

- 2021 Jamali, S. and Mitchell, D. **"An Experimental Study of Permanently Stored Learned Clauses"** Pragmatics of SAT.
- 2019 Jamali, S. and Mitchell, D. **"Simplifying CDCL Clause Database Reduction"** International Conference on Theory and Applications of Satisfiability Testing.
- 2018 Jamali, S. and Mitchell, D. **"Centrality-Based Improvements to CDCL Heuristics"** International Conference on Theory and Applications of Satisfiability Testing.
- 2017 Jamali, S. and Mitchell, D. **"Improving SAT Solver Performance with Structure-Based Preferential Bumping."** Global Conference on Artificial Intelligence.
- 2016 Jamali, S., McBride, R. and Mitchell, D. **"Experiments with CNF Structure and Hardness."** Graph Structure and Satisfiability Testing Workshop.